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📈 Logistic Regression 📈

🎯Logistic Regression

- **supervised machine learning algorithm**
- used for **classification problems**, especially when the output has only **two possible outcomes (binary classification)**

So, the **dependent variable** (**Y**) is **binary (dichotomous)** $Y \in \{0, 1\}$
> we predict the dependent variable!!

- In **linear regression**, we **predict** the **value of Y directly**.
- Problem: The predicted value can be:
  - *Less than 0* ❌
  - *Greater than 1* ❌
- But probabilities must always lie between: $0 \leq P(Y) \leq 1$

To solve this issue, Logistic Regression **predicts** a **probability** using a special function called the **Logistic (Sigmoid) Function**

📈 Logistic (Sigmoid) Function

The logistic function converts any value into a **probability between 0 and 1.**

$$P(Y) = \frac{1}{1 + e^{-z}}$$

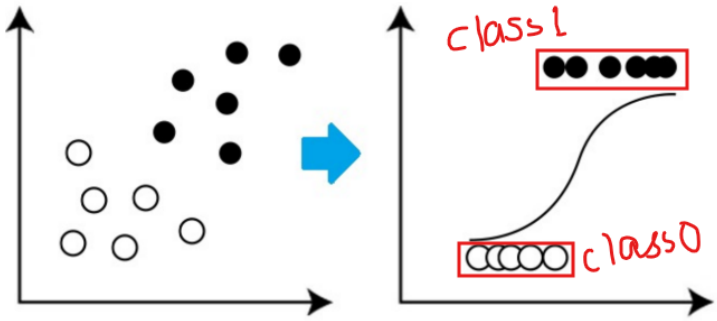
where

$$z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k$$

> Properties of Sigmoid Function

- **Output** always between **0 and 1**
- Produces an **S-shaped curve**
- Used to **estimate probabilities**

📊 Sigmoid Curve Interpretation



> Instead of drawing a straight line like Linear Regression, Logistic Regression fits an **S-shaped curve.**

x Value	Probability
Very Small	Close to 0
Medium	Around 0.5
Very Large	Close to 1

🔥 What is the "Regression" Component?

Regression means **finding the relationship between**:

- Predictor variables (X)
- Outcome variable (Y)

Logistic Regression:

- Measures association between variables
- **Predicts probabilities**
- Builds classification models

🧮 The Logistic Regression Model

The logistic regression equation is:

$$\ln\left(\frac{P(Y)}{1-P(Y)}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k$$

🎯 Outcome Variable (Y)

Y is binary, **Dichotomous Outcome** ⇒ only two possible categories.

- NO ⇒ 0
- YES ⇒ 1

Examples:

- Fail = 0, Pass = 1
- Healthy = 0, Sick = 1

📥 Predictor Variables

- The **X values** are the **input features.** $X_1, X_2, \ldots, X_k$
- These variables help predict Y.

🎯 Intercept (β_0)

- Also called: Bias OR Constant term
- the **starting value** of the model **when all X values are zero**.

🎯 Coefficients (β_1, β_2, ... β_k)

- **How much each predictor affects the outcome.**

› Interpretation

If β is:

✅ **Positive** → **increases probability**

❌ **Negative** → **decreases** probability

🎲 Odds and Log Odds

Probability of success: **P(Y)**
Example: P(Y)=0.8 ⇒ 80% chance of success.

📌 Odds › compare success against failure.

$$Odds = \frac{P(Y)}{1 - P(Y)}$$

$$Odds = \frac{0.8}{0.2} = 4$$ **Success is 4 times more likely than failure.**

📌 Log Odds (Logit)

Taking natural log of odds:

$$Logit(P) = \ln\left(\frac{P(Y)}{1 - P(Y)}\right)$$

This is exactly what Logistic Regression models.

🔄 Why Model Log Odds Instead of Probability?

Because probabilities → **restricted between 0 and 1**

But **log odds**: $-\infty \text{ to } +\infty$, This **allows** a **linear equation to work**.

🔁 Converting Log Odds Back to Probability

Starting from:

$$\ln\left(\frac{P(Y)}{1 - P(Y)}\right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_k X_k$$

Let

$$z = \beta_0 + \beta_1 X_1 + \cdots + \beta_k X_k$$

Then:

$$P(Y) = \frac{e^z}{1 + e^z}$$ or equivalently: $$P(Y) = \frac{1}{1 + e^{-z}}$$

^ This is the **final prediction equation**.

Feature	Linear Regression	Logistic Regression
Output	**Continuous** value	**Probability**
Used For	**Prediction**	**Classification**
Range	$-\infty$ to $+\infty$	0 to 1
Equation	$Y = \beta_0 + \beta_1 X + \varepsilon$	$\text{logit}(P) = \beta_0 + \beta_1 X$
Curve	**Straight** line	**S-curve**
Target Variable	**Continuous**	**Binary**

🤝 Commonality Between Linear and Logistic Regression

Both models:

- ✅ **Use predictors (X1, X2, … ,Xk)**
- ✅ **Learn coefficients β**
- ✅ Have an **intercept β_0**
- ✅ Find **relationships between variables**
- ✅ Belong to the **Generalized Linear Model (GLM)** family

> Generalized Linear Model **extends linear regression** to **different types of outputs.**

⚙️ Logistic Regression Training Algorithm

The model **learns by repeatedly adjusting weights.**

- Calculate the prediction $\hat{y}$ using the current parameters (W and b)
- Calculate the loss of the current values
- Calculate the gradients of the loss function with respect to the parameters
- Adjust the weights (optimize) using the gradients
- Repeat for the number of epochs, i.e. the number of times to go through the provided examples (dataset)

Step 1: Initialize Parameters

Randomly choose:

- Weights (W)
- Bias (b)

Step 2: Calculate Prediction

Compute: $z = WX + b$

Apply sigmoid:

$$\hat{y} = \frac{1}{1 + e^{-z}}$$ > gives **predicted probability.**

Step 3: Calculate Loss

Measure prediction error.

> Binary Cross-Entropy Loss

$$L = -\left[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})\right]$$

Step 4: Calculate Gradients

Find:

$\frac{\partial L}{\partial W}$ and $\frac{\partial L}{\partial b}$

These tell us how much to adjust parameters.

Step 5: Update Weights

Gradient Descent:

$$W = W - \alpha \frac{\partial L}{\partial W} \qquad b = b - \alpha \frac{\partial L}{\partial b}$$

Alpha = learning rate

Step 6: Repeat

Repeat Steps 2–5 for many iterations (epochs).

Each iteration:

- Loss decreases
- Predictions improve
- Model learns patterns

🧠 Logistic Regression as a Neural Network

Logistic Regression

$z = b + a_1x_1 + a_2x_2 + a_3x_3$
$p = 1.0 / (1.0 + e^{-z})$

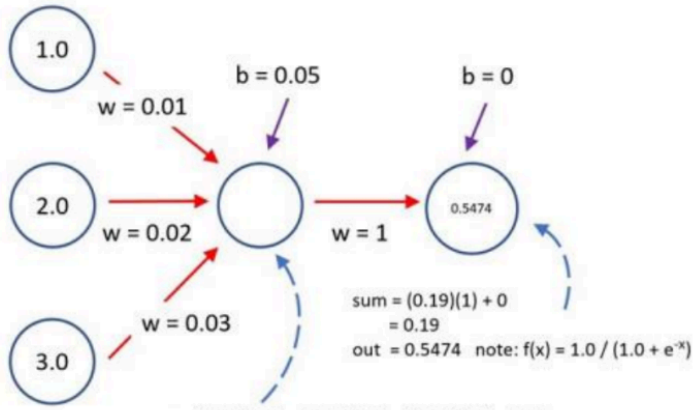
Ex:

$x_1 = 1.0$ $a_1 = 0.01$
$x_2 = 2.0$ $a_2 = 0.02$
$x_3 = 3.0$ $a_3 = 0.03$
$b = 0.05$

$z = (0.05) + (0.01)(1.0) + (0.02)(2.0) + (0.03)(3.0)$
$= 0.05 + 0.01 + 0.04 + 0.09$
$= 0.19$

$p = 1.0 / (1.0 + e^{-0.19})$
$= 0.5474$ (predicted class = 1)

Neural Network
single hidden layer, identity activation f(x) = x
single output node, logistic sigmoid activation f(x) = $1 / (1 + e^{-x})$



🎯 Loss Function

A **Loss Function** tells us **how much error the model made** for one training example.

- Small loss for **good predictions**
- Large loss for **bad predictions**

❌ Why Not Use Squared Error?

In Linear Regression we use: $L = (y - \hat{y})^2$

Problem

For Logistic Regression:Using squared error with the sigmoid function creates a:
Non-Convex Cost Function

Meaning:

- **Many local minimums**
- **Difficult optimization**
- **Gradient Descent may get stuck**

❌ Hard to find best solution (global optimum) & to minimize cost value

🎯 Solution: Cross Entropy Loss

Logistic Regression uses: **Cross Entropy Loss**

Also called:

- Log Loss
- Logistic Loss

This gives us a:

- ✅ Convex optimization problem
- ✅ Easier training
- ✅ Better convergence

$$\text{If} \quad y = 1: \qquad p(y|x) = \hat{y}$$
$$\text{If} \quad y = 0: \qquad p(y|x) = 1 - \hat{y}$$

$$P(y|x) = \hat{y}^{y} \, (1\text{-}\hat{y})^{(1\text{-}y)}$$

$$\log(P(y|x)) = \log \hat{y}^{y} \, (1\text{-}\hat{y})^{(1\text{-}y)}$$

$$= y \log \hat{y} * (1\text{-}y)(1\text{-}\hat{y})$$

🎯 What is Log Likelihood?

Likelihood ⇒ **"How likely are the observed data given the current model?"**

We want this value to be: **As Large As Possible**

Goal > Maximize Log Likelihood

> Machine Learning algorithms usually: **Minimize** instead of maximize.

So we simply multiply by: **- 1**

📉 Cross Entropy Loss

The loss for one training example becomes:

$$L_{CE} = -\left[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y}) \right]$$

Cross Entropy:

- **Rewards correct predictions**
- **Punishes** wrong predictions **heavily**

📊 Loss for Entire Dataset

For one example:

$$L_{CE} = -\left[y \log(\hat{y}) + (1-y) \log(1-\hat{y})\right]$$

For m training examples:

$$J(w,b) = \frac{1}{m} \sum_{i=1}^{m} L\left(\hat{y}^{(i)}, y^{(i)}\right)$$

Substituting the loss:

$$J(w,b) = -\frac{1}{m} \sum_{i=1}^{m} \left[y^{(i)} \log \hat{y}^{(i)} + (1-y^{(i)}) \log(1-\hat{y}^{(i)})\right]$$

This is called the **Cost Function**

🎯 Goal of Training

Find: w and b such that **J(w,b)** becomes **as small as possible.**

⚙️ Gradient Descent

- Used to **minimize J(w,b)**

🔄 Gradient Descent Algorithm [same algo as above]

> Step 1

Initialize:

- Weights (w)
- Bias (b)

with random values.

> Step 2

Compute prediction: $\hat{y} = \sigma(w^T x + b)$

> Step 3

Compute cost: **J(w,b)**

> Step 4

Compute gradients

$\frac{\partial J}{\partial w}$ & $\frac{\partial J}{\partial b}$

› Step 5

$$w = w - \alpha \frac{\partial J}{\partial w} \qquad b = b - \alpha \frac{\partial J}{\partial b}$$

Alpha ⇒ learning rate

› Step 6

Repeat Steps 2–5 for many iterations.

Each iteration:

- ✅ Cost decreases
- ✅ Predictions improve
- ✅ Parameters become optimal

Logistic Regression

- Used for binary classification.
- Predicts probabilities between 0 and 1.
- Uses the sigmoid (logistic) function.
- Models the **log odds (logit)** of the outcome.
- Output variable is **dichotomous (0 or 1).**
- β_0 is intercept.
- $\beta_1, \beta_2, \ldots, \beta$□ are coefficients.
- Belongs to the Generalized Linear Model (GLM) family.
- Trained using **gradient descent** and **cross-entropy loss.**
- Final prediction is obtained using a threshold (usually 0.5).

Loss Function

- Measures prediction error.
- Squared error is not preferred in Logistic Regression.

- Squared error creates a non-convex optimization problem.
- Logistic Regression uses Cross Entropy Loss.
- Cross Entropy is derived from Log Likelihood.
- Log Likelihood is maximized.
- Cross Entropy Loss is minimized.

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example:

- Calculate the probability of pass students who studied 33 hours.
- At least how many hours student should study that makes he will pass the course with the probability of more than 95%.

- Given the model suggested by the optimizer for odds of passing the course is,

Log(odds)=-64+2*hours

Hours Study	Pass (1) / Fail (0)
29	0
15	0
33	1
28	1
39	1

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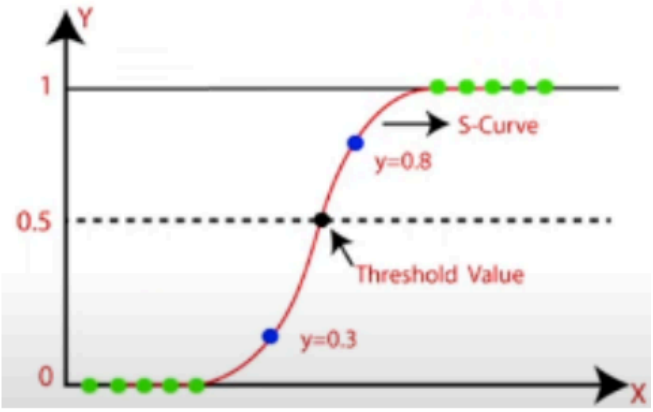
- We use Sigmoid Function in logistic regression

$$s(x) = \frac{1}{1+e^{-x}}$$



- Calculate the probability of pass students who studied 33 hours.

$$\log(odds) = z = -64 + 2 * hours$$

- So, hours= 33 which make the equation like

$$s(x) = \frac{1}{1+e^{-x}}$$

Log(odds)=z=-64+2*33

Z=2

$$p = \frac{1}{1+e^{-z}}$$

By putting the value to find the probability, we get 0.88

That is 88% chances student will pass the exam if he study 33%

- At least how many hours student should study that makes he will pass the course with the probability of more than 95%.

$$p = \frac{1}{1+e^{-z}} = 0.95$$

$$0.95 * (1 + e^{-z}) = 1$$

$$0.95 * e^{-z} = 1 - 0.95$$

$$e^{-z} = \frac{0.05}{0.95} = 0.0526$$

$$\ln(e^{-z}) = \ln(0.0526)$$

$$\ln(e^x) = x$$

$$-z = ln(0.0526) = -2.94$$

$$z = 2.94$$

$$\log(odds) = z = -64 + 2 * hours$$

$$2.94 = -64 + 2 * hours$$

$$2 * hours = 2.94 + 64$$

$$2 * hours = 66.94$$

$$hours = \frac{66.94}{2}$$

$$hours = 33.47\ Hours$$